Open-Source Pilot Proposal & Tokenized Franchise Architecture

Project Title

EcoMining AI – Global Open-Source Framework for Autonomous Tailings Remediation, Critical Metal Recovery, and RWA Franchise Tokenization

Executive Summary

Industrial mining legacies have generated thousands of un-reclaimed tailings ponds globally, creating severe environmental liabilities (acid rock drainage, toxic eolian dust) alongside unexploited reserves of critical technology metals (Tellurium, Germanium, Gallium, Rare Earths).

This project establishes a **scalable, open-source, and tokenized pilot framework** designed for global deployment. By combining closed-loop hydrometallurgical processing, an IoT/AI environmental monitoring stack, and an **RWA (Real-World Asset) tokenization mechanism ($ECO9)**, the framework provides an end-to-end blueprint. It enables municipalities, private operators, or international entities anywhere in the world to deploy local remediation projects or integrate as a **standardized franchise model**.

1. Technological Stack & Pilot Infrastructure

1.1. Modular Hydrometallurgical Processing Plant

- **Dry-Stacking & Zero-Discharge:** Eliminates wet tailings hazards by converting toxic slurries into dry, chemically inert geopolymer blocks suitable for construction or soil capping.
- **Closed-Loop Chemical Circuit:** Reagents (e.g., eco-friendly leaching agents or oxidation catalysts) operate inside sealed stainless-steel loops with 100% water recirculation, preventing runoff into local aquifers.
- **Critical Mineral Extraction:** Sequential separation targeting free gold, silver, copper, and technology metals (Tellurium, Gallium, Germanium) contained within historical residues.

1.2. IoT & Edge AI Environmental Monitoring

- **Perimeter Sensor Array:** Solar-powered LoRaWAN/5G nodes measuring particulate matter ($PM_{2.5}$, PM_{10}), wind speed/direction, soil pH, and groundwater conductivity.
- **Automated Suppression:** IoT relays trigger misting cannons with organic crusting agents when wind speed or dust thresholds exceed safety parameters.
- **Edge Computing Node:** On-site industrial GPU edge servers running local inference for automated valve controls, chemical dosing optimization, and hazard prevention without relying on external internet connectivity.

2. $ECO9 Tokenization & Web3 Financing Architecture

To solve capital allocation barriers without relying on traditional banking debt or slow government grants, the project utilizes an **RWA Tokenization Mechanism**.



2.1. Real-World Asset (RWA) Backing & Real Yield

1. **Capital Mobilization:** Initial CapEx for site equipment and IoT deployment is raised globally via $ECO9 token sales.
2. **Revenue Distribution (Real Yield):**
    - **30% Net Profit:** Allocated to on-chain **Buyback & Burn** smart contracts, driving deflationary pressure.
    - **40% Net Profit:** Distributed in stablecoins (USDT/USDC) directly to token holders staking $ECO9.
    - **30% Net Profit:** Reinvested into local site maintenance, site capping, and franchise expansion.
3. **Proof-of-Decontamination (PoD):** Real-time IoT sensor metrics (processed volume, air quality indices, recovered metal metrics) are anchored on-chain via Chainlink Oracles to provide immutable verification for ESG investors.

3. Global Franchise & Open-Source Licensing Model

3.1. Franchise Blueprint for Global Entities

Any regional authority, enterprise, or community group facing local environmental contamination can adopt the **EcoMining AI Franchise**:

- **Turnkey Deployment Package:** Standardized CAD schematics for modular plants, pre-configured IoT sensor kits, and edge computing containers.
- **Franchise Royalty & Staking Integration:** Franchisees stake $ECO9 tokens as a performance bond, receiving access to centralized AI models, automated compliance reporting tools, and global buyer networks for recovered critical metals.

3.2. Open-Source Philosophy

If commercial franchise routes are unavailable or restricted by local bureaucracy, the entire core codebase (IoT sensor ingestion, AI chemical control loops, and Web3 smart contracts) is released under the **MIT / Apache 2.0 Open-Source License**. This ensures any community globally can freely build, adapt, and run their own remediation infrastructure.

4. Python Source Code: Open-Source AI Monitoring Engine

Below is the standalone Python core engine (ecomining_core.py). It ingests IoT sensor feeds, uses machine learning logic to predict dust dispersion risks, controls automated suppression cannons, and generates on-chain data payloads for the $ECO9 smart contract interface.

Python

```
import time
import json
import random
import hashlib
from datetime import datetime

class IoTSensorArray:
    """Simulates or reads physical IoT nodes deployed around tailings site."""
    def __init__(self, node_id):
        self.node_id = node_id

    def read_telemetry(self):
        # In production, replace with actual serial/LoRaWAN/MQTT sensor reads
        return {
            "node_id": self.node_id,
            "timestamp": datetime.utcnow().isoformat(),
            "pm2_5": round(random.uniform(5.0, 150.0), 2),  150 µg/m³ = high alert
            "pm10": round(random.uniform(10.0, 300.0), 2),
            "wind_speed_kmh": round(random.uniform(0.0, 65.0), 1),
            "ph_level": round(random.uniform(2.5, 8.5), 2),  < 4.0 = Acid Rock Drainage
            "groundwater_ec": round(random.uniform(200, 2500), 1)  # µS/cm
        }

class EcoMiningAIEngine:
    """Core AI Inference engine for environmental safety & dosing control."""
    def __init__(self, pm_threshold=75.0, wind_threshold=35.0):
        self.pm_threshold = pm_threshold
        self.wind_threshold = wind_threshold
        self.suppression_active = False

    def evaluate_risk(self, telemetry):
        """Predictive risk evaluation logic for dust suppression and chemical dosing."""
        pm2_5 = telemetry["pm2_5"]
        wind_speed = telemetry["wind_speed_kmh"]
        ph = telemetry["ph_level"]

        actions = []
        risk_score = 0.0

        # Dust Dispersion Risk Assessment
        if pm2_5 > self.pm_threshold or wind_speed > self.wind_threshold:
            risk_score += 0.6
            actions.append("ACTIVATE_MIST_CANNONS")
            self.suppression_active = True
        else:
```

```
            self.suppression_active = False

        # Acid Rock Drainage Risk Assessment
        if ph < 4.0:
            risk_score += 0.4
            actions.append("INCREASE_ALKALINE_NEUTRALIZER_DOSING")

        status = "CRITICAL" if risk_score >= 0.6 else "NOMINAL"

        return {
            "status": status,
            "risk_score": round(risk_score, 2),
            "recommended_actions": actions,
            "suppression_system": "ON" if self.suppression_active else "OFF"
        }

class ProofOfDecontaminationOracle:
    """Generates immutable hashes for Web3 / $ECO9 smart contract state updates."""
    @staticmethod
    def generate_pod_payload(telemetry, evaluation, processed_tonnes):
        payload = {
            "telemetry": telemetry,
            "evaluation": evaluation,
            "processed_tonnes_cumulated": processed_tonnes,
            "nonce": random.randint(100000, 999999)
        }
        serialized = json.dumps(payload, sort_keys=True).encode('utf-8')
        pod_hash = hashlib.sha256(serialized).hexdigest()

        return {
            "payload": payload,
            "pod_hash": f"0x{pod_hash}",
            "oracle_status": "READY_FOR_ONCHAIN_SUBMISSION"
        }

# Main Execution Loop (Pilot Test Simulation)
if __name__ == "__main__":
    print("=== EcoMining AI: Open-Source Engine Initialized ===")
    sensor_node = IoTSensorArray(node_id="NODE-MOLDOVA-NOUA-001")
    ai_engine = EcoMiningAIEngine(pm_threshold=75.0, wind_threshold=35.0)

    simulated_processed_tonnes = 1250.50

    try:
        for cycle in range(1, 4):
            print(f"\n--- Execution Cycle #{cycle} ---")
            data = sensor_node.read_telemetry()
            analysis = ai_engine.evaluate_risk(data)
            simulated_processed_tonnes += 15.25

            oracle_packet = ProofOfDecontaminationOracle.generate_pod_payload(
                data, analysis, round(simulated_processed_tonnes, 2)
            )

            print(f"[IoT Raw Feeds]  PM2.5: {data['pm2_5']} µg/m³ | Wind: {data['wind_speed_kmh']} km/h |
pH: {data['ph_level']}")
```

```
            print(f"[AI Decision]   Status: {analysis['status']} | Actions: {analysis['recommended_actions']}")
            print(f"[Proof-of-Decontamination Hash] {oracle_packet['pod_hash']}")

            time.sleep(2)
    except KeyboardInterrupt:
        print("\nShutdown signal received. Engine stopped safely.")
```

5. Budget & Implementation Blueprint (Pilot Site)

CAPEX & Deployment Schedule (24-Month Pilot Phase)

Category	Cost (EUR)
Modular Closed-Loop Plant & High-Pressure Filters	€ 4,800,000
IoT Sensor Field Mesh, Edge AI Hardware & Drones	€ 1,000,000
Open-Source Code Base Audit & Smart Contract Security	€ 350,000
Operational Staffing (24 Months)	€ 1,500,000
Re-vegetation, Site Capping & Environmental Audits	€ 1,200,000
Contingency Reserve (10%)	€ 885,000
TOTAL INITIAL CAPEX	**€ 9,735,000**

6 .EcoMining AI: Global Open-Source Framework & RWA Tokenization Architecture ($ECO999)

6.1. Executive Summary & Open-Source Franchise Concept

Industrial legacy mining sites (such as Moldova Nouă and Baia Mare) represent severe environmental hazards characterized by toxic eolian dust and acid rock drainage (ARD). At the same time, they contain unexploited reserves of critical technology metals (Tellurium, Germanium, Gallium) and precious metals.

This framework provides a **turnkey open-source blueprint and franchise model** that can be deployed globally. It merges closed-loop hydrometallurgical processing, an IoT/Edge AI environmental monitoring mesh, and an **RWA (Real-World Asset) tokenization mechanism**

($ECO999) to bypass traditional banking barriers and fund site remediation directly through global decentralized finance.

6.2. Detailed RWA Tokenization Process ($ECO999)

Tokenizing a real-world mining and ecological restoration project bridges physical industrial assets with blockchain liquidity through a transparent, audited legal structure:

Step 1: Legal Structuring & SPV Formation

- **Special Purpose Vehicle (SPV):** A dedicated corporate entity (SPV) is established in a Web3-friendly and EU-compliant jurisdiction (e.g., Switzerland, Luxembourg, or Cyprus).
- **Asset Backing:** Concession rights, processing infrastructure, and future revenue streams from recovered critical metals are legally tied to the SPV.
- **MiCA Compliance:** Prospectuses are structured in accordance with European **MiCA (Markets in Crypto-Assets)** regulations to classify the token correctly as an Asset-Referenced or Utility/Revenue-Share Token.

Step 2: Smart Contract Deployment & Security Audits

- **Code Architecture:** Smart contracts (deployed on EVM-compatible networks like Polygon or Arbitrum for low gas fees) codify immutable rules: a hard cap of token supply, staking logic, and automated revenue-sharing algorithms.
- **Security Audits:** Independent cybersecurity firms audit the smart contract code to eliminate vulnerabilities or backdoors.

Step 3: IoT & Oracle Data Synchronization (Proof-of-Decontamination)

- **On-Chain Transparency:** Industrial IoT sensors and plant scales transmit encrypted hourly data feeds via decentralized oracles (e.g., Chainlink).
- **Proof-of-Decontamination (PoD):** Every kilogram of processed material and recovered metal is hashed onto the blockchain, giving ESG investors cryptographic proof of real-world environmental impact.

Step 4: Capital Raising (CAPEX Funding)

- **Token Distribution:** The **$ECO999** token is offered via private rounds (ESG/crypto funds) and public launchpads to raise initial capital for modular plant procurement and IoT sensor deployment.

Step 5: Real Yield & Buyback Mechanics

- **Revenue Flow:** Critical metals extracted from the tailings are sold to industrial buyers.
- **Smart Contract Execution:**

- **40% Net Profit** is distributed automatically in stablecoins (USDT/USDC) to token holders staking $ECO999.
- **30% Net Profit** executes automated **Buyback & Burn** smart contracts, reducing circulating supply and creating deflationary pressure.

6.3. International Tokenization Firms

The following global companies provide turnkey infrastructure for legally issuing and managing RWA tokenization projects:

1. **Securitize (USA / Switzerland):** A global leader in tokenizing real-world assets and investment funds, ensuring full KYC/AML compliance and secondary market trading.
2. **Centrifuge (Germany / Switzerland):** Pioneers in bridging real-world assets to decentralized finance (DeFi), enabling liquidity pools for industrial CapEx financing.
3. **Tokeny Solutions (Luxemburg):** Provides end-to-end tokenization infrastructure fully compliant with EU regulations (ERC-3643 standard) for MiCA-aligned issuance.
4. **Polymath / Polymesh (Canada / Switzerland):** Specializes in building institutional-grade, regulated blockchain infrastructure tailored for mineral rights, assets, and franchise models.
5. **Fireblocks (USA / Israel):** Enterprise-grade digital asset custody and treasury management platform securing project funds and smart contract interactions.

6.4. Python Source Code: ECO999 Tokenization & Real-Yield Engine

Below is the standalone Python script modeling the **$ECO999** tokenization mechanics, simulating IoT sensor feeds, metal recovery revenues, automated Buyback & Burn protocols, and staking distributions.

Python

```
import time
import json
import random
import hashlib
from datetime import datetime

class ECO999TokenomicsEngine:
    """Simulates $ECO999 Token economics, real-yield distribution, and IoT
oracles."""
    def __init__(self, total_supply=100_000_000):
        self.token_name = "ECO999"
        self.total_supply = total_supply
        self.circulating_supply = 40_000_000  # Initial public/private sale
pool
        self.staked_tokens = 15_000_000
        self.treasury_fund_eur = 0.0

    def simulate_industrial_output(self):
```

```
        """Simulates weekly metal recovery (Tellurium, Gold, Copper) sold to
industrial buyers."""
        tonnes_processed = round(random.uniform(800.0, 1500.0), 2)
        revenue_eur = round(tonnes_processed * random.uniform(45.0, 85.0), 2)
        net_profit_eur = round(revenue_eur * 0.65, 2)  # 65% net margin after
OPEX
        return tonnes_processed, net_profit_eur

    def execute_smart_contract_distribution(self, net_profit_eur):
        """Executes automated real-yield allocation and buyback-burn mechan-
ics."""
        staking_yield_pool = net_profit_eur * 0.40  # 40% distributed to
stakers (USDT/USDC)
        buyback_burn_pool = net_profit_eur * 0.30   # 30% used to buyback and
burn ECO999
        retained_ops_pool = net_profit_eur * 0.30   # 30% retained for site
expansion

        # Simulate deflationary burn
        tokens_burned = int(buyback_burn_pool / 2.50)  # Assuming token price
= €2.50
        self.circulating_supply -= tokens_burned

        return {
            "net_profit_generated": net_profit_eur,
            "staker_dividend_pool_eur": round(staking_yield_pool, 2),
            "buyback_burn_eur": round(buyback_burn_pool, 2),
            "eco999_tokens_burned": tokens_burned,
            "new_circulating_supply": self.circulating_supply
        }

class ProofOfDecontaminationOracle:
    """Generates cryptographic proof of environmental remediation for on-
chain verification."""
    @staticmethod
    def generate_pod_hash(cycle_data):
        payload = json.dumps(cycle_data, sort_keys=True).encode('utf-8')
        h = hashlib.sha256(payload).hexdigest()
        return f"0x{h}"

# Main Execution Loop (Simulation)
if __name__ == "__main__":
    print(f"=== ECO999 RWA Tokenization & Decontamination Engine Initialized
===")
    engine = ECO999TokenomicsEngine()

    try:
        for week in range(1, 4):
            print(f"\n--- Operational & Token Cycle #{week} ---")
            tonnes, profit = engine.simulate_industrial_output()
            distribution = engine.execute_smart_contract_distribution(profit)

            cycle_record = {
                "timestamp": datetime.utcnow().isoformat(),
                "tonnes_processed": tonnes,
                "net_profit": profit,
                "distribution": distribution
```

```
            }

            pod_hash = ProofOfDecontaminationOracle.generate_pod_hash(cy-
cle_record)

            print(f"[Industrial Output] Processed: {tonnes} tonnes | Net
Profit: €{profit:,.2f}")
            print(f"[Smart Contract]    Staking Pool: €{distribu-
tion['staker_dividend_pool_eur']:,.2f}")
            print(f"[Deflationary Burn] Burned: {distribution['eco999_to-
kens_burned']:,} ECO999 tokens")
            print(f"[Circulating Supply]: {distribution['new_circulating_sup-
ply']:,} ECO999")
            print(f"[On-Chain Oracle PoD Hash]: {pod_hash}")

            time.sleep(2)
    except KeyboardInterrupt:
        print("\nSimulation stopped safely.")
```

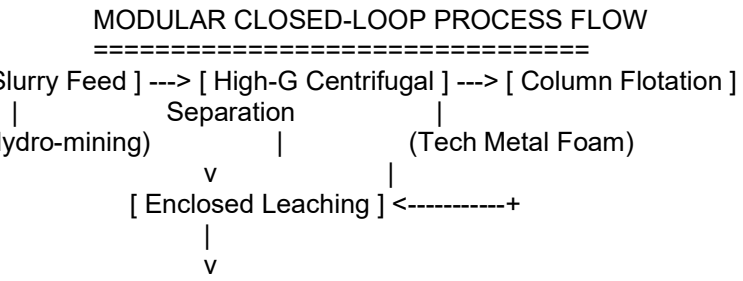
To complete the technical proposal package for international environmental authorities, investors, and technical auditors, here is the detailed technical specification section in English.

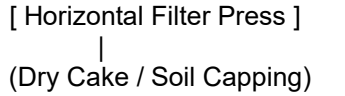
This section covers the **Modular Plant engineering details**, the **IoT and Edge AI operational matrix**, and an **International Source & Vendor Registry** listing established global suppliers where auditors can verify project viability at the source.

Technical Supplement: Modular Plant Architecture, IoT/AI System Specs, and Vendor Verification Registry

1. Modular Processing Plant Engineering Details

The modular processing plant is engineered for rapid deployment, scalability, and strict compliance with global zero-discharge (*BAT – Best Available Techniques*) standards.



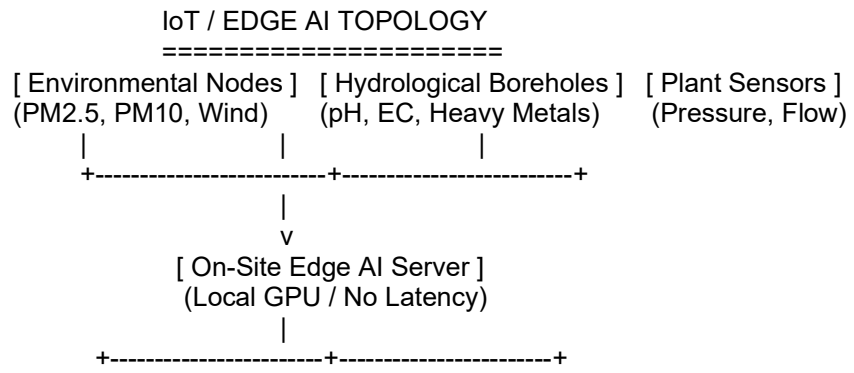


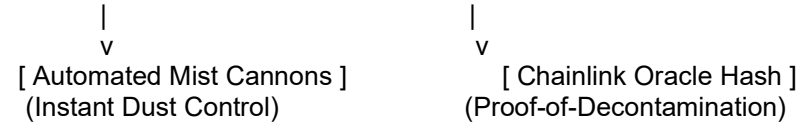
1.1. Core Module Specifications

- **Containerized Skid Design:** All processing units are mounted on standardized ISO skid frames (20ft / 40ft containers) allowing rapid transport by road or rail and installation on lightweight concrete pads without permanent landscape destruction.
- **Hydro-Mining & Slurry Feed Module:** High-pressure water monitors (up to 25 bar) slurry the tailings in-situ. Heavy-duty slurry pumps (high-chrome alloy wet ends) transport the material through HDPE pipelines at flow rates of $150 - 400$ m^3/h.
- **High-G Fine Grinding & Centrifugal Separation Module:**
  - **Stirred Media Detritors (SMD):** Milled down to fine grain sizes ($P_{80} < 25$ μm) to liberate encapsulated metallic minerals.
  - **Knelson / Falcon Concentrators:** Centrifugal force fields ($60 - 200$ G) capture free gold, silver, and heavy mineral complexes without chemicals.
- **Enclosed Leaching & Neutralization Reactors:**
  - Sealed 316L stainless-steel pressurized reactors equipped with mechanical agitators and dissolved-oxygen injectors.
  - Inline catalytic oxidation loop (SO_2/Air or Hydrogen Peroxide) destroys any residual leaching agents in a closed feedback loop before filtration.
- **High-Pressure Horizontal Membrane Filter Press Module:**
  - Operating pressures up to 16 bar reduce cake moisture below 15%.
  - Produces chemically inert, dry-stacked "cakes" used for structural backfill or capped with organic topsoil.
  - **100% Process Water Recycling:** All squeezed liquid is returned to the slurry headquarter pumps.

2. IoT Mesh & Edge AI Operational System Architecture

The digital layer acts as an automated "nervous system" and "brain" operating across the site.





2.1. Industrial IoT (Internet of Things) Infrastructure

- **Environmental Perimeter Mesh:** Solar-powered LoRaWAN nodes equipped with laser particulate sensors ($PM_{2.5}$, PM_{10}), ultrasonic anemometers (wind vector/velocity), and humidity gauges.
- **Hydro-Geochemical Subsurface Monitoring Ring:** Multiparameter borehole probes measuring pH, oxidation-reduction potential (ORP), electrical conductivity (EC), and dissolved heavy-metal ions in surrounding aquifers.
- **Plant Process IoT:** Inline magnetic flow meters, radiometric density meters, and vibration sensors transmitting telemetry every 500 milliseconds.

2.2. Edge Artificial Intelligence (AI) & Software Stack

- **Advanced Process Control (APC) Machine Learning Engine:**
  - Neural network models process real-time pulp density and mineralogy to regulate reagent pumps with millimetric accuracy, preventing chemical overdosage and optimizing metal recovery.
- **Predictive Environmental Risk Model:**
  - Integrates real-time weather radar data with perimeter IoT feeds to predict dust dispersion up to 72 hours in advance. Automatically triggers mist-cannon arrays using eco-binding agents if wind speeds exceed critical thresholds.
- **Computer Vision & Thermal Analytics:**
  - Thermal-imaging cameras mounted on automated drone stations scan tailings surfaces 24/7 to identify self-heating chemical hotspots or structural slope anomalies.

3. Global Source & Vendor Registry for Project Verification

To verify the viability of this project, technical auditors and environmental guards can cross-reference equipment specs, software systems, and IoT hardware directly with established global vendors:

A. Modular Mineral Processing & Filtration Equipment

1. **Metso (Finland):** Global leader in modular grinding mills, column flotation cells, and high-pressure horizontal filter presses (*Metso VP-Series*).
   - *Website / Source:* metso.com
2. **FLSmidth (Denmark):** World-class supplier of eco-friendly hydrometallurgical circuits, slurry pumping systems, and dry-stack tailings filtration.
   - *Website / Source:* flsmidth.com

3. **Eriez Magnetics / HydroFloat (USA):** Industry-standard fine-mineral separation and high-efficiency column flotation technology.
    - *Website / Source:* eriez.com

B. IoT Hardware, Sensor Mesh & Environmental Automation

1. **Libelium (Spain / EU):** Specialist in industrial Smart Environment IoT sensor networks (*Waspmote / Plug & Sense!* arrays for $PM_{2.5}$, PM_{10}, pH, and water quality).
    - *Website / Source:* libelium.com
2. **Endress+Hauser (Switzerland):** Global manufacturer of inline chemical sensors, pH/EC probes, and electromagnetic flow meters for heavy industrial processes.
    - *Website / Source:* endress.com
3. **Campbell Scientific (USA / UK):** Automated weather stations, geotechnical slope-stability sensors, and hydrological logging systems.
    - *Website / Source:* campbellsci.com

C. Edge AI Hardware, Process Control Software & Oracles

1. **NVIDIA Enterprise (USA):** Industrial Edge AI computing hardware (e.g., *NVIDIA Jetson AGX Industrial* / *Nvidia IGX*) for containerized on-site inference without cloud dependency.
    - *Website / Source:* nvidia.com/en-us/edge-computing/
2. **Schneider Electric / AVEVA (France / UK):** Advanced Process Control (APC) and SCADA software platforms integrated with machine-learning optimization engines for mineral processing.
    - *Website / Source:* se.com / aveva.com
3. **Chainlink Labs (USA / Global):** Decentralized Oracle network providing the software middleware to bridge IoT sensor telemetry directly to blockchain smart contracts (*Proof-of-Decontamination*).
    - *Website / Source:* chain.link